					Course Code: B23CT4101
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R23**
IV B.Tech. I Semester MODEL QUESTION PAPER					
FUNDAMENTALS OF DEEP LEARNING					
(Common to CSD & CSIT)					
Time: 3 Hrs.					**Max. Marks: 70 M**
Answer Question No.1 compulsorily					
Answer **ONE Question** from **EACH UNIT**					
Assume suitable data if necessary					
					10 X 2=20 Marks

			CO	KL	M
1	**a)**	Define **Deep Learning**.	**1**	**1**	**2**
	b)	What is meant by **model complexity** in deep learning?	**1**	**1**	**2**
	c)	What is a **Perceptron**?	**2**	**1**	**2**
	d)	What is the role of **bias input** in a perceptron?	**2**	**1**	**2**
	e)	Difference between the **batch training and sequential training**.	**3**	**2**	**2**
	f)	What is meant by **stochastic gradient descent (SGD)**?	**3**	**1**	**2**
	g)	Define a **Convolutional Neural Network (CNN)**.	**4**	**1**	**2**
	h)	Define **pooling** in convolutional networks.	**4**	**1**	**2**
	i)	Define a **Bidirectional RNN**.	**5**	**1**	**2**
	j)	What is meant by **explicit memory** in sequence modeling?	**5**	**1**	**2**
					5 x 10 = 50 Marks
		UNIT-I			
2.	**a)**	Explain how **deep learning models achieve higher accuracy** compared to traditional machine learning techniques.	**1**	**2**	**5**
	b)	Describe the relationship between **complexity and performance** in deep learning models.	**1**	**2**	**5**
		OR			
3.	**a)**	Discuss the **real-world impact of Deep Learning** in various application domains.	**1**	**2**	**5**
	b)	Discuss the challenges and future scope of **Deep Learning technologies**.	**1**	**2**	**5**
		UNIT-II			
4.	**a)**	Explain the structure and working of the **biological neuron and artificial neuron**.	**2**	**2**	**5**
	b)	Explain the concept of **Linear Regression** and compare it with perceptron-based learning.	**2**	**2**	**5**
		OR			
5.	**a)**	Explain the **Perceptron Learning Algorithm** with a suitable example.	**2**	**2**	**5**
	b)	Define **Neural Networks** and explain their characteristics and applications.	**2**	**2**	**5**

		UNIT-III			
6.	**a)**	Explain the architecture and working of a **Multi-Layer Perceptron (MLP)** with a neat diagram.	**3**	**2**	**5**
	b)	Describe the process of **initializing weights** in neural networks and its importance.	**3**	**2**	**5**
		OR			
7.	**a)**	Explain the importance of the **amount of training data and number of hidden layers** in MLP performance.	**3**	**2**	**5**
	b)	Discuss different criteria used to decide **when to stop learning** in neural network training.	**3**	**2**	**5**
		UNIT-IV			
8.	**a)**	Describe the **convolution operation** and explain how feature maps are generated.	**4**	**2**	**5**
	b)	Explain different types of **pooling techniques** used in CNNs.	**4**	**2**	**5**
		OR			
9.	**a)**	Discuss the role of **random or unsupervised features** in convolution networks.	**4**	**2**	**5**
	b)	Explain the **neuro scientific basis for convolution networks** and its influence on deep learning models.	**4**	**2**	**5**
		UNIT-V			
10.	**a)**	Explain the concept of **sequence modelling** and its importance in deep learning applications.	**5**	**2**	**5**
	b)	Explain the **Encoder–Decoder Sequence-to-Sequence architecture** and its applications.	**5**	**2**	**5**
		OR			
11.	**a)**	Discuss the **challenge of long-term dependencies** in RNNs and methods to overcome them.	**5**	**2**	**5**
	b)	Describe the architecture and advantages of **LSTM and other gated RNNs**. Explain the role of **explicit memory** in sequence modelling.	**5**	**2**	**5**

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: B23HS4101
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
HUMAN RESOURCE & PROJECT MANAGEMENT					
(Common to CSE, AIML, IT, AIDS, CSG, CSIT)					
Time: 3 Hrs.					**Max. Marks: 70 M**
Answer Question No.1 compulsorily					
Answer **ONE Question** from **EACH UNIT**					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	**KL**	**M**
1.	**a).**	Write the role of HR manager.	1	1	2
	b).	Define recruitment.	1	1	2
	c).	List any two models of HR accounting.	2	1	2
	d).	Define career counseling.	2	1	2
	e).	Define Resource management.	3	1	2
	f).	State the 80-20 rule.	3	1	2
	g).	Define Brainstorming.	4	1	2
	h).	Write about Cash flow analysis.	4	1	2
	i).	List any two forms of project organization.	5	1	2
	j).	Define project control.	5	1	2
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Interpret the nature and scope of Human Resource Management.	1	2	5
	b).	Discuss the emerging trends in HRM and their impact on organizations.	1	2	5
		OR			
3.	**a).**	Explain the ethical aspects of HRM with suitable examples.	1	2	5
	b).	Describe the steps involved in the selection procedure.	1	2	5
		UNIT-2			
4.	**a).**	Discuss the different methods of training used in organizations.	2	2	5
	b).	Explain the importance of performance appraisal in organizations.	2	2	5
		OR			
5.	**a).**	Summarize the concept of career development and its importance.	2	2	5
	b).	Explain group interaction and its importance in HRD.	2	2	5
		UNIT-3			
6.	**a).**	Explain the concept of Project Management and its importance.	3	2	5
	b).	Discuss the different types of projects with examples.	3	2	5
		OR			

7.	**a).**	Explain the stages of the project life cycle.	**3**	**2**	**5**
	b).	Interpret the process of project selection.	**3**	**2**	**5**
		UNIT-4			
8.	**a).**	Describe the criteria used for project selection.	**4**	**2**	**5**
	b).	A company invests ₹2,00,000 in a project. The expected cash inflows are: Year 1: ₹50,000 Year 2: ₹60,000 Year 3: ₹70,000 Year 4: ₹80,000 Calculate the Payback Period.	**4**	**2**	**5**
		OR			
9.		A project requires ₹2,00,000 investment. It generates ₹80,000 annually for 4 years. Discount rate = 10% Calculate the NPV and state whether the project is acceptable.	**4**	**2**	**10**
		UNIT-5			
10.	**a).**	Interpret the process of project planning in detail.	**5**	**2**	**5**
	b).	Explain the human aspects of project management.	**5**	**2**	**5**
		OR			
11.	**a).**	Discuss the prerequisites for successful project implementation.	**5**	**2**	**5**
	b).	Summarize the importance of project review.	**5**	**2**	**5**

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

					Course Code: B23CT4102
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R23**
		IV B.Tech. I Semester MODEL QUESTION PAPER			
		INFORMATION SECURITY			
		(For CSIT)			
Time: 3 Hrs.					**Max. Marks: 70 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					10 X 2=20 Marks
			CO	**KL**	**M**
1	**a)**	What is meant by **Role-Based Access Control (RBAC)**?	**1**	**1**	**2**
	b)	Define **Identity and Access Management (IAM)**.	**1**	**1**	**2**
	c)	Discuss **computer virus**?	**2**	**2**	**2**
	d)	What is meant by a **root kit**?	**2**	**1**	**2**
	e)	List any two methods used to defend against **buffer overflow attacks**.	**3**	**1**	**2**
	f)	Explain the **safe program code**?	**3**	**2**	**2**
	g)	Describe a **Denial-of-Service (DoS) attack**.	**4**	**2**	**2**
	h)	What is the purpose of a **Firewall**?	**4**	**1**	**2**
	i)	Discuss the **Wireless Security**?	**5**	**2**	**2**
	j)	Explain the **Cloud Security**?	**5**	**2**	**2**
					5 x 10 = 50 Marks
		UNIT-I			
2.	**a)**	Explain the principles and importance of **Digital User Authentication**.	**1**	**2**	**5**
	b)	Discuss various **security issues related to user authentication**.	**1**	**2**	**5**
		OR			
3.	**a)**	Describe the **UNIX File Access Control mechanism** with examples.	**1**	**2**	**5**
	b)	Demonstrate the concept and methods of Remote User Authentication**.**	**1**	**3**	**5**
		UNIT-II			
4.	**a)**	Explain the concepts of **backdoors and rootkits** and their impact on system security.	**2**	**2**	**5**
	b)	Illustrate the concept and characteristics of an Advanced Persistent Threat (APT).	**2**	**3**	**5**
		OR			
5.	**a)**	Explain the need and importance of Database and Data Center Security**.**	**2**	**2**	**5**
	b)	Illustrate different types of information theft attacks such as key loggers, phishing, and spyware.	**2**	**3**	**5**
		UNIT-III			

6.	**a)**	Discuss the causes and consequences of Stack Overflow Attacks with suitable examples.	**3**	**2**	**5**
	b)	Explain different methods for handling program input securely.	**3**	**2**	**5**
		OR			
7.	**a)**	Describe the common programming errors that result in software security issues.	**3**	**2**	**5**
	b)	Examine the different techniques used for defending against Buffer Overflow attacks.	**3**	**3**	**5**
		UNIT-IV			
8.	**a)**	Identify the different types of **Flooding Attacks** with suitable examples.	**4**	**2**	**5**
	b)	Demonstrate the concept and importance of Intrusion Detection Systems (IDS) in network security.	**4**	**3**	**5**
		OR			
9.	**a)**	Explain the characteristics and access policies of a Firewall.	**4**	**2**	**5**
	b)	Illustrate the concept and components of Public-Key Infrastructure (PKI).	**4**	**3**	**5**
		UNIT-V			
10.	**a)**	Illustrate the major Cloud Security concepts and challenges in cloud environments	**5**	**3**	**5**
	b)	Interpret the various security threats and challenges in IoT systems.	**5**	**3**	**5**
		OR			
11.	**a)**	Explain the overview of Wireless Security and its importance in modern communication systems.	**5**	**2**	**5**
	b)	Demonstrate the common wireless network attacks and preventive measures.	**5**	**3**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

					Course Code: B23AD4102
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R23**
		IV B.Tech. I Semester MODEL QUESTION PAPER			
		Devops			
		Common to AIDS & CSIT			
Time: 3 Hrs.					**Max. Marks: 70 M**
		Answer Question No.1 compulsorily			
		Answer **ONE** Question from **EACH UNIT**			
		Assume suitable data if necessary			
					10 x 2 = 20 Marks
			CO	**KL**	**M**
1.	**a).**	What is the purpose of SDLC in software development?	**1**	**2**	**2**
	b).	How does DevOps improve the software development process?	**1**	**2**	**2**
	c).	What is the need for version control systems?	**2**	**2**	**2**
	d).	Why are branches used in Git?	**2**	**2**	**2**
	e).	What is the main goal of Continuous Integration?	**3**	**2**	**2**
	f).	What is the role of Jenkins in CI?	**3**	**2**	**2**
	g).	Why are Docker containers used in application deployment?	**4**	**2**	**2**
	h).	What is the purpose of Continuous Delivery?	**4**	**2**	**2**
	i).	What is the use of Ansible in DevOps?	**5**	**2**	**2**
	j).	What is the role of a Pod in Kubernetes?	**5**	**2**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Explain the phases of SDLC with suitable examples.	**1**	**2**	**5**
	b).	Explain DevOps lifecycle and the role of automation in continuous improvement.	**1**	**2**	**5**
		OR			
3.	**a).**	Explain Agile Model and Compare Scrum and Kanban frameworks	**1**	**3**	**5**
	b).	Illustrate DevOps architecture and its key components.	**1**	**3**	**5**
		UNIT-2			
4.	**a).**	Explain the need for Source Code Management (SCM). Describe the evolution of SCM systems leading to Git.	**2**	**2**	**5**
	b).	Explain the Git workflow, clearly describing the roles of working directory, staging area, local repository, and remote repository.	**2**	**2**	**5**
		OR			
5.	**a).**	Explain Git branching and collaboration in team environments	**2**	**2**	**5**
	b).	Describe Unit Testing and explain the features of JUnit and NUnit frameworks used for automated testing.	**2**	**2**	**5**

		UNIT-3			
6.	**a).**	Explain Build Automation and its importance in DevOps.	**3**	**2**	**5**
	b).	Explain Jenkins Architecture and Master–Slave model. Describe its working with a diagram.	**3**	**2**	**5**
		OR			
7.	**a).**	Explain Jenkins workflow with steps.	**3**	**2**	**5**
	b)	Compare Freestyle Projects and Jenkins Pipelines.	**3**	**3**	**5**
		UNIT-4			
8.	**a).**	Explain the importance of Continuous Delivery in DevOps.	**4**	**2**	**5**
	b)	Explain Dockerfile instructions and working of containers along with its lifecycle	**4**	**2**	**5**
		OR			
9.	**a).**	Explain Continuous Deployment flow with diagram.	**4**	**2**	**5**
	b)	Define Docker. Explain Docker Images, Containers, Commands and Dockerfile.	**4**	**2**	**5**
		UNIT-5			
10.	**a).**	Explain Ansible architecture and key features.	**5**	**2**	**5**
	b).	Explain Kubernetes architecture, namespaces, and resources.	**5**	**3**	**5**
		OR			
11.	**a).**	Explain Ansible roles and Jinja templating with examples.	**5**	**2**	**5**
	b).	Explain CI/CD in OpenShift. Describe BuildConfig (BC), DeploymentConfig (DC), and ConfigMaps	**5**	**3**	**5**

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: B23CT4103
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R23**
		IV B.Tech. I Semester MODEL QUESTION PAPER			
		DATA SCIENCE USING PYTHON			
		(Common to CSIT & CSD)			
Time: 3 Hrs.					**Max. Marks: 70 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					10 X 2=20 Marks
			CO	**KL**	**M**
1	**a)**	State the concept of Data Science.	**1**	**1**	**2**
	b)	Outline the stages involved in the Data Science life cycle.	**1**	**1**	**2**
	c)	What is the purpose of feature selection.	**2**	**1**	**2**
	d)	Identify supervised learning with a suitable example.	**2**	**2**	**2**
	e)	Distinguish unsupervised learning from supervised learning.	**3**	**2**	**2**
	f)	Summarize Big Data and list any two characteristics.	**3**	**2**	**2**
	g)	State the role of Hadoop in Big Data processing.	**4**	**1**	**2**
	h)	Discuss the significance of NoSQL databases.	**4**	**2**	**2**
	i)	Explain the need for data preprocessing.	**5**	**2**	**2**
	j)	Illustrate the importance of data visualization in Data Science.	**5**	**3**	**2**
					5 x 10 = 50 Marks
		UNIT-I			
2.	**a)**	Explain the complete Data Science process with a neat diagram.	**1**	**2**	**10**
		OR			
3.	**a)**	Summarize the concepts and characteristics of Big Data.	**1**	**3**	**5**
	b)	Discuss the major challenges involved in Data Science.	**1**	**2**	**5**
		UNIT-II			
4.	**a)**	Explain the concept and significance of feature selection in Machine Learning.	**2**	**2**	**5**
	b)	Discuss different techniques used for handling missing data.	**2**	**2**	**5**
		OR			
5.	**a)**	Classify the types of Machine Learning and demonstrate each with suitable examples.	**2**	**3**	**10**
		UNIT-III			
6.	**a)**	Examine the Hadoop framework and explain the functions of its components.	**3**	**3**	**10**
		OR			
7.	**a)**	Interpret the working process of Map Reduce with a suitable example.	**3**	**3**	**5**

	b)	Demonstrate the advantages of NoSQL databases in Big Data applications.	**3**	**3**	**5**
		UNIT-IV			
8.	**a)**	Demonstrate various data visualization tools and techniques used in Python.	**4**	**3**	**10**
		OR			
9.	**a)**	Examine the features and applications of Mat plot lib or Sea born	**4**	**3**	**5**
	b)	Demonstrate any one real-time application of Data Science.	**4**	**3**	**5**
		UNIT-V			
10.	**a)**	Compare Artificial Intelligence, Machine Learning, and Deep Learning with suitable examples.	**5**	**4**	**10**
			5		**5**
		OR			
11.	**a)**	Examine the role and applications of Natural Language Processing (NLP).	**5**	**3**	**5**
	b)	Demonstrate any one real-time application of Artificial Intelligence such as chatbots or recommendation systems.	**5**	**3**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23CT4104

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A) **R23**

IV B.Tech. I Semester MODEL QUESTION PAPER

MOBILE COMPUTING

(For CSIT)

Time: 3 Hrs. **Max. Marks: 70 M**

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 X 2=20 Marks

			CO	KL	M
1	**a).**	Which frequency range is suitable for satellite communication and why?	**1**	**1**	**2**
	b).	Explain the wireless transmission concepts to Wi-Fi communication.	**1**	**2**	**2**
	c).	Demonstrate the use of TDMA in GSM communication.	**2**	**3**	**2**
	d).	Illustrate the applications of telecommunication systems in modern industries.	**2**	**3**	**2**
	e).	Demonstrate the difference between infrared and radio transmission.	**3**	**3**	**2**
	f).	Apply Bluetooth communication in smart home applications.	**3**	**3**	**2**
	g).	Compare traditional IP routing and Mobile IP in handling mobility.	**4**	**4**	**2**
	h).	Explain the impact of packet loss on Traditional TCP performance.	**4**	**2**	**2**
	i).	Explain WAP concepts used in mobile banking applications.	**5**	**2**	**2**
	j).	Demonstrate the role of WTLS in mobile banking applications.	**5**	**3**	**2**
					5 x 10 = 50 Marks
		UNIT-I			
2.	**a).**	Illustrate how the evolution of wireless networks has influenced the development of modern mobile communication systems.	**1**	**3**	**5**
	b).	Describe the simplified reference model of wireless communication in a mobile internet application.	**1**	**2**	**5**
		OR			
3.	**a).**	Illustrate the use of Frequency Division Multiplexing (FDM) and Time Division Multiplexing (TDM) in practical wireless communication systems.	**1**	**3**	**5**
	b).	Demonstrate frequency management techniques to allocate channels efficiently in a densely populated city	**1**	**3**	**5**
		UNIT-II			
4.	**a).**	Examine how SDMA can be applied in a crowded cellular network to improve spectrum utilization and reduce interference.	**2**	**3**	**5**
	b).	Interpret how CDMA technology enables multiple users to communicate simultaneously using unique spreading codes.	**2**	**3**	**5**
		OR			

5.	**a).**	Explain how GSM subscriber and service data management maintain user profiles and services in a network.	**2**	**3**	**5**
	b).	Demonstrate how UMTS improves data transmission rates compared to earlier GSM systems.	**2**	**K3**	**5**
		UNIT-III			
6.	**a).**	Illustrate the differences between infrared and radio transmission using suitable wireless communication applications.	**3**	**3**	**5**
	b).	What is the use of radio transmission in WLANs and its advantages in mobile communication environments.	**3**	**1**	**5**
		OR			
7.	**a).**	Discuss the how adhoc networking enables peer-to-peer communication without fixed infrastructure.	**3**	**2**	**5**
	b).	Implement the WLAN services in ensuring secure and reliable wireless access.	**3**	**3**	**5**
		UNIT-IV			
8.	**a).**	Examine the working of Mobile IP and analyze how it maintains communication continuity when a mobile node changes its network location.	**4**	**4**	**5**
	b).	Compare infrared and radio transmission in suitable wireless communication applications.	**4**	**4**	**5**
		OR			
9.	**a).**	Illustrate the performance challenges of TCP during handoff situations in cellular networks	**4**	**3**	**5**
	b).	Evaluate different TCP improvement approaches used in mobile communication systems and compare their effectiveness.	**4**	**4**	**5**
		UNIT-V			
10.	**a).**	Explain how Wireless Application Protocol (WAP) enables mobile devices to access web-based applications efficiently.	**5**	**2**	**5**
	b).	Interpret the role of different layers in the WAP architecture during wireless communication.	**5**	**3**	**5**
		OR			
11.	**a).**	Demonstrate support for mobility concepts in maintaining communication continuity for roaming wireless users.	**5**	**3**	**5**
	b).	Explain how wireless application environment concepts support multimedia services in mobile communication systems.	**5**	**2**	**5**

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: B23CT4105
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R23**
IV B.Tech. I Semester MODEL QUESTION PAPER					
CYBER SECURITY					
(For CSIT)					
Time: 3 Hrs.					**Max. Marks: 70 M**
Answer Question No.1 compulsorily					
Answer **ONE Question** from **EACH UNIT**					
Assume suitable data if necessary					
					10 X 2=20 Marks
			CO	**KL**	**M**
1	**a)**	Define Cybercrime and its categories.	1	1	2
	b)	Explain Botnet? and its role in cybercrime?	1	2	2
	c)	What is Phishing? how phishing attacks are conducted?	2	2	2
	d)	Differentiate between DoS and DDoS attacks.	2	3	2
	e)	What is Digital Evidence?	3	1	2
	f)	Explain E-mail Tracking and its role in cybercrime investigation	3	2	2
	g)	What is Computer Forensics?	4	1	2
	h)	List any four computer forensics software tools and their purpose.	4	1	2
	i)	Describe the key provisions of the Indian IT Act.	5	2	2
	j)	What are Digital Signatures?	5	1	2
					5 x 10 = 50 Marks
		UNIT-I			
2.		Explain the different classifications of Cybercrime and its security challenges posed by mobile devices.	1	2	10
		OR			
3.	**a)**	Define cyber stalking and its types, discuss how individuals can protect themselves.	1	1	5
	b)	Illustrate different types of network and computer attacks, and describe attack vectors with examples.	1	3	5
		UNIT-II			
4.		Explain SQL Injection and Buffer Overflow attacks in detail.	**2**	**2**	10
		OR			
5.	**a)**	Interpret the Password Cracking and Keylogging methods.	**2**	**3**	**5**
	b)	Compare Steganography and Social Engineering. How are they used to facilitate cybercrimes?	**2**	**4**	**5**
		UNIT-III			
6.	**a)**	Explain the steps involved in digital evidence collection and evidence preservation with a practical scenario.	**3**	**2**	**5**

	b)	Apply E-mail Investigation and IP Tracking procedures used in cybercrime investigation.	**3**	**3**	**5**
		OR			
7.	**a)**	Illustrate the search and seizure of computers and the process of recovering deleted evidence	**3**	**3**	**5**
	b)	Describe Encryption and Decryption Methods used in evidence analysis. Discuss password cracking in investigation.	**3**	**2**	**5**
		UNIT-IV			
8.		Explain the procedure for preparing a computer forensics investigation, and also its role in the forensics hardware and software tools.	4	2	10
		OR			
9.	**a)**	Differentiate between the Windows System Forensics and Linux System Forensics.	4	3	5
	b)	Discuss the Cell Phone and Mobile Device Forensics and challenges.	4	2	5
		UNIT-V			
10.	**a)**	Analyze the Indian IT Act provisions and its amendments.	**5**	4	**5**
	b)	Explain Digital Signatures under the Indian IT Act their legal validity.	**5**	2	**5**
		OR			
11.	**a)**	Compare the cybercrime legal frameworks of India and other countries.	**5**	**4**	**5**
	b)	Discuss cybercrime punishment under Indian law. Analyze notable cybercrime cases and their legal outcomes.	**5**	**2**	**5**

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: B23CT4107
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R23**
IV B.Tech. I Semester MODEL QUESTION PAPER					
DATA ENGINEERING					
(For CSIT)					
Time: 3 Hrs.					**Max. Marks: 70 M**
Answer Question No.1 compulsorily					
Answer **ONE Question** from **EACH UNIT**					
Assume suitable data if necessary					
					10 X 2=20 Marks
			CO	**KL**	**M**
1	a)	Explain the Data Engineering Lifecycle?	**1**	**2**	**2**
	b)	List any four skills required for a Data Engineer**.**	**1**	**1**	**2**
	c)	What is meant by Enterprise Architecture**?**	**2**	**1**	**2**
	d)	Discuss the source systems in data engineering?	**2**	**2**	**2**
	e)	What is meant by eventual consistency?	**3**	**1**	**2**
	f)	Discuss the concept of streaming storage.	**3**	**2**	**2**
	g)	Difference between the batch ingestion and stream ingestion**?**	**4**	**1**	**2**
	h)	State the Importance of message queues and event-streaming platforms**?**	**4**	**1**	**2**
	i)	Explain materialized view**.**	**5**	**2**	**2**
	j)	List any two ways to serve data for analytics and machine learning.	**5**	**1**	**2**
					5 x 10 = 50 Marks
		UNIT-I			
2.	**a)**	Discuss the evolution of Data Engineering and its growing significance in industry.	**1**	**2**	**5**
	b)	Explain the roles and responsibilities of Data Engineers inside an organization**.**	**1**	**2**	**5**
		OR			
3.	**a)**	Discuss the processes of data storage, ingestion, transformation, and serving data**.**	**1**	**2**	**5**
	b)	Differentiate between the Data Lifecycle and the Data Engineering Lifecycle**.**	**1**	**2**	**5**
		UNIT-II			
4.	**a)**	Explain the concept of Enterprise Architecture and its relationship with Data Architecture.	**2**	**2**	**5**
	b)	Describe the roles and responsibilities of people involved in designing a data architecture**.**	**2**	**2**	**5**
		OR			
5.	**a)**	Explain the concepts of data sharing and third-party data sources in modern data systems.	**2**	**2**	**5**

	b)	Discuss the role and importance of message queues and event-streaming platforms in data engineering.	**2**	**2**	**5**
		UNIT-III			
6.	**a)**	Difference between the eventual consistency and strong consistency.	**3**	**2**	**5**
	b)	Describe various storage abstractions used in data engineering**.**	**3**	**2**	**5**
		OR			
7.	**a)**	Differentiate between single-machine storage and distributed storage systems**.**	**3**	**2**	**5**
	b)	Discuss the concept of separation of compute from storage and its advantages.	**3**	**2**	**5**
		UNIT-IV			
8.	**a)**	Describe the role of message queues and event-streaming platforms in data ingestion.	**4**	**2**	**5**
	b)	Explain the practical issues associated with common file formats during data ingestion.	**4**	**2**	**5**
		OR			
9.	**a)**	Explain the concept and importance of Data Ingestion in data engineering.	**4**	**2**	**5**
	b)	Discuss different data ingestion techniques such as webhooks, web interfaces, web scraping, and transfer appliances for data migration.	**4**	**2**	**5**
		UNIT-V			
10.	**a)**	Explain the role and importance of queries in data engineering systems.	**5**	**2**	**5**
	b)	Discuss the importance of streaming transformations and stream processing in real-time analytics.	**5**	**2**	**5**
		OR			
11.	**a)**	Desctribe the role of a Data Engineer in Machine Learning applications**.**	**5**	**2**	**5**
	b)	Explain different types of data transformations with suitable examples.	**5**	**2**	**5**

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: B23CT4108
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R23**
IV B.Tech. I Semester MODEL QUESTION PAPER					
ARTIFICIAL INTELLIGENCE AND NEURAL NETWORKS					
(For CSIT)					
Time: 3 Hrs.					**Max. Marks: 70 M**
Answer Question No.1 compulsorily					
Answer **ONE Question** from **EACH UNIT**					
Assume suitable data if necessary					
					10 X 2=20 Marks
			CO	**KL**	**M**
1	**a)**	What is Soft Computing? Differentiate it from Hard Computing.	1	2	2
	b)	List and describe the steps of A* search algorithm.	1	2	2
	c)	Explain the McCulloch-Pitts neuron model.	2	2	2
	d)	Define Backpropagation. State the learning rule used in it.	2	2	2
	e)	Define Fuzzy Set and Membership Function with an example.	3	2	2
	f)	What is Defuzzification? List any two methods.	3	1	2
	g)	State the basic components of a Genetic Algorithm.	4	2	2
	h)	What is Particle Swarm Optimization? Describe its main operators.	4	2	2
	i)	What is a Neuro-Fuzzy system? State its purpose.	5	2	2
	j)	Explain the concept of ANFIS briefly.	5	2	2
					5 x 10 = 50 Marks
		UNIT-I			
2.		Explain the A* and AO* search algorithms with suitable examples. Compare their performance.	1	2	10
		OR			
3.	**a)**	Describe the different types of Soft Computing Techniques and their applications.	1	2	5
	b)	Explain Knowledge Representation using Semantic Networks and Frames with examples.	1	2	5
		UNIT-II			
4.		Explain the Backpropagation Neural Network algorithm in detail with its training procedure, convergence conditions and limitations.	**2**	**2**	10
		OR			
5.	**a)**	Compare Single Layer and Multilayer Perceptrons. Discuss the Perceptron Learning Rule.	**2**	**2**	**5**
	b)	Describe Hopfield Network architecture and explain its working with an example.	**2**	**2**	**5**

		UNIT-III			
6.	**a)**	Define Fuzzy Sets and explain Membership Functions. Illustrate with Union, Intersection and Complement operations.	**3**	**2**	**5**
	b)	Explain Mamdani Fuzzy Inference System with all stages: fuzzification, rule evaluation, aggregation and defuzzification.	**3**	**2**	**5**
		OR			
7.	**a)**	Explain Fuzzy Rule-Based Models and Linguistic Variables with an example of a fuzzy controller.	**3**	**2**	**5**
	b)	Describe the Sugeno Fuzzy Inference System. How does it differ from Mamdani? Illustrate with an example.	**3**	**2**	**5**
		UNIT-IV			
8.		Explain the working of a Genetic Algorithm with a suitable example covering encoding, selection, crossover and mutation operators.	4	2	10
		OR			
9.	**a)**	Explain Particle Swarm Optimization (PSO) algorithm. Describe position and velocity update equations.	4	2	5
	b)	Describe Ant Colony Optimization (ACO). Apply it to the Travelling Salesman Problem.	4	2	5
		UNIT-V			
10.	**a)**	Explain the architecture and working of ANFIS (Adaptive Neuro-Fuzzy Inference System). State its learning procedure.	**5**	**2**	**5**
	b)	Describe Genetic Algorithm Based Backpropagation Network. How does GA help in optimizing weights?	**5**	**2**	**5**
		OR			
11.	**a)**	Explain Neuro-Fuzzy Hybrid Systems with a block diagram. List its advantages over standalone systems.	**5**	**2**	**5**
	b)	Describe Genetic-Fuzzy Hybrid Systems. State any two real-world application areas.	**5**	**2**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

		Course Code: B23CT4109			
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R23**
		IV B.Tech. I Semester MODEL QUESTION PAPER			
		SNOW FLAKE CLOUD ANALYSIS			
		(For CSIT)			
Time: 3 Hrs.					**Max. Marks: 70 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					10 X 2=20 Marks
			CO	**KL**	**M**
1	**a)**	Define cloud data warehousing.	**1**	**2**	**2**
	b)	What is Snowflake architecture?	**1**	**2**	**2**
	c)	What is Snowpipe?	**2**	**2**	**2**
	d)	Define stages in Snowflake.	**2**	**2**	**2**
	e)	What is VARIANT data type?	**3**	**2**	**2**
	f)	Define data flattening.	**3**	**2**	**2**
	g)	What is RBAC?	**4**	**2**	**2**
	h)	Define masking policy.	**4**	**2**	**2**
	i)	What is query caching?	**5**	**2**	**2**
	j)	Define resource monitor.	**5**	**2**	**2**
					5 x 10 = 50 Marks
		UNIT-I			
2.	**a)**	Explain cloud data warehousing and its advantages.	**1**	**2**	**5**
	b)	Explain Snowflake architecture and its components.	**1**	**2**	**5**
		OR			
3.	**a)**	Describe Snowflake storage, compute, and cloud services layers.	**1**	**2**	**5**
	b)	Explain virtual warehouses and database objects in Snowflake.	**1**	**2**	**5**
		UNIT-II			
4.	**a)**	Illustrate data loading process using stages and COPY command.	**2**	**3**	**5**
	b)	Apply SQL queries (joins, aggregations) in Snowflake with example.	**2**	**3**	**5**
		OR			
5.	**a)**	Demonstrate Snowpipe for automatic data ingestion.	**2**	**3**	**5**
	b)	Apply views and temporary tables in query processing.	**2**	**3**	**5**
		UNIT-III			
6.	**a)**	Illustrate querying semi-structured data using VARIANT with example.	**3**	**3**	**5**
	b)	Apply streams and tasks for change data capture.	**3**	**3**	**5**
		OR			

7.	**a)**	Demonstrate ETL/ELT data transformation techniques.	**3**	**3**	**5**
	b)	Apply stored procedures and UDFs in Snowflake.	**3**	**3**	**5**
		UNIT-IV			
8.	**a)**	Analyze RBAC mechanism in securing Snowflake data.	**4**	**4**	**5**
	b)	Analyze data sharing techniques in Snowflake.	**4**	**4**	**5**
		OR			
9.	**a)**	Analyze encryption and masking policies for data protection.	**4**	**4**	**5**
	b)	Analyze auditing and monitoring mechanisms for governance.	**4**	**4**	**5**
		UNIT-V			
10.	**a)**	Apply query performance optimization techniques.	**5**	**3**	**5**
	b)	Demonstrate warehouse scaling and caching mechanisms.	**5**	**3**	**5**
		OR			
11.	**a)**	Apply monitoring techniques to track Snowflake usage.	**5**	**3**	**5**
	b)	Demonstrate cost optimization using resource monitors.	**5**	**3**	**5**

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: B23CT4110
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R23**
		IV B.Tech. I Semester MODEL QUESTION PAPER			
		ADVANCE COMPUTING			
		(For CSIT)			
Time: 3 Hrs.					**Max. Marks: 70 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					10 X 2=20 Marks
			CO	**KL**	**M**
1	**a)**	Define Grid Computing.	**1**	**1**	**2**
	b)	Define Autonomic Computing.	**1**	**1**	**2**
	c)	What are commodity components?	**2**	**1**	**2**
	d)	What is load sharing?	**2**	**1**	**2**
	e)	Define Pervasive Computing.	**3**	**1**	**2**
	f)	What is context-aware computing?	**3**	**1**	**2**
	g)	What is quantum measurement?	**4**	**1**	**2**
	h)	Define uncertainty principle.	**4**	**2**	**2**
	i)	Define Hilbert space	**5**	**1**	**2**
	j)	What is a Hermitian operator?	**5**	**1**	**2**
					5 x 10 = 50 Marks
		UNIT-I			
2.	**a)**	Explain the architecture of Grid Computing.	**1**	**2**	**5**
	b)	Explain the role of schedulers and resource brokers	**1**	**3**	**5**
		OR			
3.	**a)**	Discuss the components of Grid infrastructure.	**1**	**2**	**5**
	b)	Compare Grid Computing with other distributed technologies.	**1**	**4**	**5**
		UNIT-II			
4.	**a)**	Discuss different classifications of clusters.	**2**	**2**	**5**
	b)	Explain flexible load sharing algorithm.	**2**	**3**	**5**
		OR			
5.	**a)**	Describe applications of cluster computing	**2**	**2**	**5**
	b)	Explain how scalable services are constructed in clusters.	**2**	**3**	**5**
		UNIT-III			
6.	**a)**	Explain the concepts and scenarios of pervasive computing.	**3**	**2**	**5**
	b)	Explain human-machine interface in pervasive systems.	**3**	**3**	**5**
		OR			
7.	**a)**	Explain Java applications in pervasive devices.	**3**	**3**	**5**

	b)	Discuss real-world applications of pervasive computing.	**3**	**2**	**5**
		UNIT-IV			
8.	**a)**	Explain the concept of superposition with examples.	**4**	**2**	**5**
	b)	Describe quantum measurement in multiple bases.	**4**	**3**	**5**
		OR			
9.	**a)**	Explain the BB84 quantum cryptography protocol.	**4**	**3**	**5**
	b)	Analyze differences between classical bits and qubits.	**4**	**4**	**5**
		UNIT-V			
10.	**a)**	Discuss linear operators and matrices.	**5**	**2**	**5**
	b)	Explain Hilbert spaces with Dirac notation.	**5**	**3**	**5**
		OR			
11.	**a)**	Analyze properties of Hermitian operators.	**5**	**4**	**5**
	b)	Describe quantum states and observables.	**5**	**2**	**5**

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CD4107

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A) | **R23**

IV B.Tech. I Semester MODEL QUESTION PAPER

BIG DATA ANALYTICS

(Common to CSD & CSIT)

Time: 3 Hrs. | **Max. Marks: 70 M**

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 X 2=20 Marks

			CO	**KL**	**M**
1	**a)**	Define Big Data and list any four applications of Big Data Analytics.	**1**	**2**	**2**
	b)	Explain the purpose of wrapper classes in Hadoop serialization.	**1**	**2**	**2**
	c)	List the major differences between GFS and HDFS.	**2**	**2**	**2**
	d)	Explain the significance of replication factor in HDFS.	**2**	**2**	**2**
	e)	Describe the role of shuffle and sort phase in Map Reduce.	**3**	**2**	**2**
	f)	Explain the function of Record Reader in Map Reduce.	**3**	**2**	**2**
	g)	List the responsibilities of Mapper and Reducer with suitable examples.	**4**	**2**	**2**
	h)	Explain the working principle of FM Algorithm in stream processing.	**4**	**2**	**2**
	i)	List the features of Apache Spark	**5**	**2**	**2**
	j)	Compare Pig Latin and HiveQL.	**5**	**2**	**2**
					5 x 10 = 50 Marks
		UNIT-I			
2.	**a)**	Explain the architecture of Hadoop Distributed File System (HDFS) and describe how it supports fault tolerance and scalability with a neat diagram.	**1**	**2**	**5**
	b)	Explain the concept of Serialization in Hadoop. Demonstrate how Writable interface is used for serialization with a suitable Java example.	**1**	**2**	**5**
		OR			
3.	**a)**	Explain the characteristics of Big Data. Discuss the challenges involved in storing and processing Big Data applications.	**1**	**2**	**5**
	b)	Explain Hadoop V1 and Hadoop V2 architectures and compare their building blocks with suitable diagrams.	**1**	**2**	**5**
		UNIT-II			
4.	**a)**	Explain different Hadoop cluster configurations namely Local mode, Pseudo-distributed mode, and Fully distributed mode with suitable examples.	**2**	**2**	**5**
	b)	Describe HDFS read and write operations with neat sketches and discuss the role of NameNode and DataNode during these operations.	**2**	**2**	**5**

		OR			
5.	**a)**	Discuss the anatomy of a Map Reduce job execution in YARN architecture with neat diagram.	**2**	**2**	**5**
	b)	Explain Rack Awareness in HDFS. How does Hadoop improve data reliability using replication and rack awareness?	**2**	**2**	**5**
		UNIT-III			
6.	**a)**	Explain the role of Combiner, Partitioner, and Record Reader in Hadoop Map Reduce framework with suitable examples.	**3**	**2**	**5**
	b)	Develop a Map Reduce program to calculate the frequency of words in a given input file and explain Mapper, Reducer, and Driver classes.	**3**	**3**	**5**
		OR			
7.	**a)**	Discuss Hadoop Streaming and explain how non-Java applications can be executed using Hadoop Streaming.	**3**	**2**	**5**
	b)	Implement Matrix Multiplication using Map Reduce and explain the processing steps involved in Mapper and Reducer phases.	**3**	**3**	**5**
		UNIT-IV			
8.	**a)**	Explain the Stream Data Model and Architecture. Discuss the importance of stream processing in real-time applications.	**4**	**2**	**5**
	b)	Illustrate the working of DGIM Algorithm for counting the number of 1's in a sliding window with a suitable example.	**4**	**3**	**5**
		OR			
9.	**a)**	Explain Apache Spark architecture and describe the functions of Spark Core, Spark SQL, Spark Streaming, MLlib, and GraphX.	**4**	**2**	**5**
	b)	Write a Spark program to demonstrate basic RDD transformations and actions. Explain the advantages of RDDs in Big Data processing.	**4**	**3**	**5**
		UNIT-V			
10.	**a)**	Explain Pig Architecture and discuss various data processing operators used in Pig Latin with suitable examples.	**5**	**2**	**5**
	b)	Illustrate the architecture of HBase and explain the role of ZooKeeper in coordinating distributed operations.	**5**	**3**	**5**
		OR			
11.	**a)**	Explain Hive Architecture and discuss different Hive data types with suitable examples.	**5**	**2**	**5**
	b)	Develop HiveQL queries to create a table, insert data, and perform filtering, grouping, and sorting operations with suitable examples.	**5**	**3**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks